MTH209: Endsem Exam - Question 3

1. Follow the instructions **exactly**.
2. If your code is not properly commented or horizontal/vertical spacing is missing, points will be deducted.
3. You are NOT allowed to use any external libraries.

Question

Suppose we observe training data consisting of predictors $X \in \mathbb{R}^p$ and class labels $Y \in \{1, \dots, K\}$. In Quadratic Discriminant Analysis (QDA), we assume that the predictors within each class follow a multivariate normal distribution,

$$X \mid Y = k \sim N(\mu_k, \Sigma_k),$$

where:
- μ_k is the mean vector for class k,
- Σ_k is the covariance matrix for class k,
- $\pi_k = P(Y = k)$ is the prior probability of class k.

Unlike LDA, QDA allows each class to have its own covariance matrix. For any observation x, the QDA discriminant score for class k is

$$\delta_k(x) = -\frac{1}{2}\log\det(\Sigma_k) - \frac{1}{2}(x - \mu_k)^\top \Sigma_k^{-1}(x - \mu_k) + \log(\pi_k).$$

Given a new observation x_{new}, QDA predicts the class label by selecting the class with the largest discriminant score:

$$\hat{y} = \arg\max_k \delta_k(x_{\text{new}}).$$

Function to be corrected

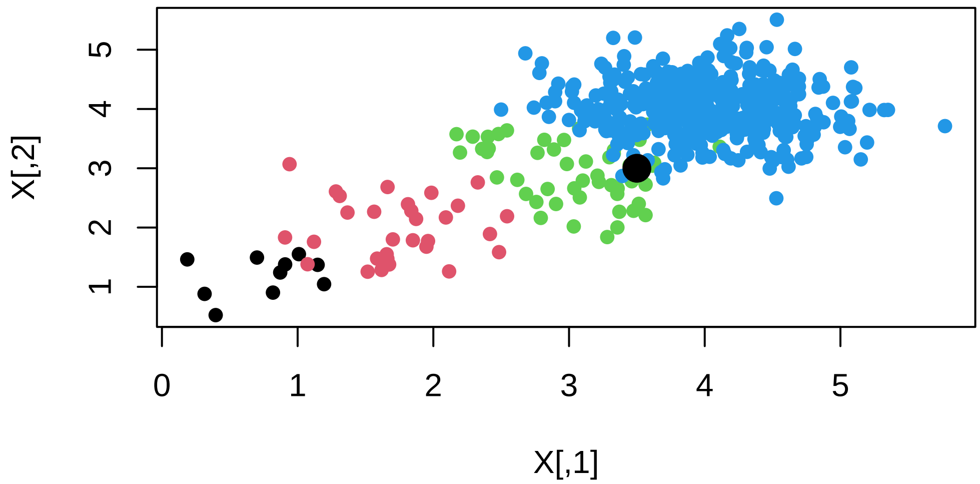
Below I have attempted to write an R function called `predict_qda()` with inputs that takes the training data and x_{new} and outputs the class prediction label for x_{new}. However, there are many errors and your job is to fix all errors.

```r
# y: a numeric vector of length n, with integer class labels
# X: an (n x p) times matrix of observations. Each row is an observation
# x_new: a numeric vector of length p
predict_qda <- function(y, X, x_new)
{
  n <- length(y)
  K <- length(y)

  # obtain mu_hat_k and Sigma_hat_k
  for(k in 1:K)
  {
```

```
    ind <- (y == k)
    yk <- y[ind]
    Xk <- X[, ind]
    pik <- sum(ind)/pik

    muk <- rowMeans(Xk)
    Sigmak <- cov(t(X))

    deltak <- -0.5*log(det(Sigmak)) - t(x_new - muk) %*% Sigmak %*% (x_new - muk) + log(pik)
  }

  return(which.min(deltak))
}
```

Below is a test code with the expected output to help you. However, please note that it is possible that your code is incorrect but the class prediction is correct.

```
set.seed(10) # don't change seed
K <- 4
n <- 500
p <- 2
ind.list <- list(1:10, 11:40, 41:90, 91:n)

y <- numeric(length = n)
X <- matrix(0, nrow = n, ncol = p)
for(k in 1:K)
{
  ind <- ind.list[[k]]
  nk <- length(ind)
  X[ind, ] <- matrix(rnorm(nk*p, mean = k, sd = .5), nrow = nk, ncol = p)
  y[ind] <- k
}
plot(X, col = y, pch = 16)
x_new <- c(3.5, 3)
points(x_new[1], x_new[2], col = "black", pch = 16, cex = 2)
```




```
predict_qda(y, X, x_new)
```

```
[1] 3
```

Submission Instructions

INSTRUCTIONS: copy and paste ONLY the `predict_qda` function and NOTHING ELSE in a file called `question3.R` and upload it to mooKIT. This question has 25 marks.